

Detailed Diagnostic Report (DDR)

1. Property Issue Summary

Property: Not Available

Address: Not Available

Property Type: Not Available

Inspection Date: Not Available

Overall Condition: Poor

Reason: The property has visible water stains, dampness, and structural cracks, indicating potential water damage and structural issues. These concerns require immediate attention and further investigation.

Total Issues Identified: 3

Key Concerns:

- Active water staining and dampness in the Master Bedroom ceiling
- Structural crack in the Basement north wall
- Inconsistent HVAC vent airflow and temperature variations in the Kitchen

2. Area-wise Observations

Master Bedroom Ceiling

[INS-001] Water stains and dampness are visible on the ceiling.

Technical Finding: Visible water stains and dampness, Cold anomaly with a temperature difference of 5.2°C

Moisture: Moderate

Source: Inspection Report, Thermal Report

Image Not Available

Basement North Wall

[INS-002] A structural crack has been found in the wall.

Technical Finding: Structural crack observed

Source: Inspection Report

Image Not Available

[THR-002] Cold spot near crack

Source: Thermal Report

Image Not Available

Kitchen HVAC Vent

[INS-003] There is a minor inconsistency in the airflow from the vent.

Technical Finding: Minor airflow inconsistency, Conflicting telemetry requiring manual verification

Source: Inspection Report, Thermal Report

Image Not Available

3. Probable Root Cause

[INS-001] Master Bedroom Ceiling: Water seepage, Active moisture infiltration causing thermal bridging

[INS-002] Basement North Wall: Settlement cracks due to soil movement

[THR-002] Basement North Wall: Not Available

[INS-003] Kitchen HVAC Vent: Not Available

4. Severity Assessment

[INS-001] Master Bedroom Ceiling - Severity: CRITICAL

Reasoning: Visible water stains and dampness indicate a significant issue, The presence of moisture and a significant temperature difference pose a significant risk

[INS-002] Basement North Wall - Severity: HIGH

Reasoning: Structural cracks are a significant concern

[THR-002] Basement North Wall - Severity: MEDIUM

Reasoning: The presence of a cold spot near a crack poses a potential risk, but the lack of thermal image data limits the assessment

[INS-003] Kitchen HVAC Vent - Severity: LOW

Reasoning: Minor inconsistency, but still worth further investigation, The inconsistency in temperature variation poses a potential issue, but the lack of clear evidence limits the assessment

5. Recommended Actions

- (CRITICAL) Water Stains and Dampness: Immediately inspect and repair the source of the water seepage, and consider hiring a professional to assess and remediate any mold growth. [Confidence: High]
- (HIGH) Structural Crack: Monitor the crack for any changes or signs of further damage, and consider hiring a structural engineer to assess the integrity of the building. [Confidence: High]
- (MEDIUM) Cold Spot Near Crack: Obtain a thermal image of the area to further assess the cold spot, and consider hiring a professional to investigate and repair any potential issues. [Confidence: Medium]
- (LOW) Minor Airflow Inconsistency: Verify the airflow inconsistency through manual measurement, and consider adjusting or cleaning the HVAC vent to improve airflow efficiency. [Confidence: Medium]

6. Additional Notes

- Water staining likely active, Relative humidity near ceiling estimated at 85%
- Basement crack requires monitoring
- Thermal image unavailable
- HVAC issue requires further verification, Kitchen readings show conflicting telemetry
- This DDR cross-references 4 observation(s) and 0 extracted image(s) from the inspection and thermal reports.

Conflicts Between Reports:

- Moisture severity conflict: Inspection Report says 'Moderate' but Thermal Report indicates a significant temperature difference of 5.2°C, suggesting 'Severe' moisture (Involves: Inspection Report, Thermal Report). Recommended verification: Verify the moisture level at the Master Bedroom Ceiling by checking for water stains, dampness, and taking precise temperature and humidity readings
- Severity assessment conflict: Inspection Report says 'High' due to structural crack, but Thermal Report says 'Medium' due to cold spot near crack, indicating a potential discrepancy in risk assessment (Involves: Inspection Report, Thermal Report). Recommended verification: Inspect the Basement North Wall to verify the severity of the structural crack and the cold spot, and assess the potential risk to the building's integrity

7. Missing or Unclear Information

- Thermal data not available for Basement North Wall observations
- Photographic evidence not available for Basement North Wall crack
- Crack width measurements not recorded for the Basement North Wall crack

- Measurements not recorded for Master Bedroom Ceiling water stains and dampness
- Root cause analysis not provided for Kitchen HVAC Vent issue
- Inspection date not mentioned in any of the observations
- Temperatures not recorded for Basement North Wall and Kitchen HVAC Vent observations
- Thermal image data not available for Basement North Wall cold spot observation